

Fixed Effect Models:

Another approach for longitudinal data

Recall Random-Effects Model

- Data: $(Y_{ij}, X_{1ij}, \mathbf{X}_{2i}, \dots, \mathbf{X}_{pi})$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$
- Random-Effects Model:
 - Level 1 model: $Y_{ij} = \beta_{0i} + \beta_{1i}X_{1ij} + \epsilon_{ij}$
 - Level 2 model: many scenarios:
 - $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1$
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 - $\beta_{i0} = \beta_0 + \beta_{02}X_{2i} + \dots + \beta_{0p}X_{pi} + b_{0i}$ and $\beta_{i1} \equiv \beta_1$,
where $b_{0i} \sim N(0, \sigma_{0b}^2)$
 - $\beta_{i0} \equiv \beta_0$ and $\beta_{i1} = \beta_1 + \beta_{12}X_{2i} + \dots + \beta_{1p}X_{pi} + b_{1i}$,
where $b_{1i} \sim N(0, \sigma_{1b}^2)$
 - $\beta_{i0} = \beta_0 + \beta_{02}X_{2i} + \dots + \beta_{0p}X_{pi} + b_{0i}$ and
 $\beta_{i1} = \beta_1 + \beta_{12}X_{2i} + \dots + \beta_{1p}X_{pi} + b_{1i}$
 - ❖ b_{0i} and b_{1i} are uncorrelated
 - ❖ b_{0i} and b_{1i} are correlated

Fixed-Effect Model

Fixed-Effects Model:

$$Y_{ij} = \beta_{0i} + \beta_1 X_{1ij} + \epsilon_{ij}$$

- All individuals are constrained to have the same slope, but have different intercepts.
- Allowing the intercepts to vary give us a (one-way) fixed effects estimate.
 - Similar ideas have also been for Cox's proportional hazards model in survival analysis.
- It is a good way to control for a long list of **unobserved** variables ($X_{2i}, \dots, X_{pi}$) that are fixed over time.

Parameter Estimates

Fixed-Effects Model:

$$Y_{ij} = \beta_{0i} + \beta_1 X_{1ij} + \epsilon_{ij}$$

- Two Methods to estimate β_1 (and β_{0i}):
 - “absorbing the fixed effects”
 - “Adding binary Indicator variable” for each subject

1. “absorbing the fixed effects”

“Absorbing the fixed effects” method to estimate β_1 :

- **Key Ideas:**

$$Y_{ij} = \beta_{0i} + \beta_1 X_{1ij} + \epsilon_{ij}$$

$$\bar{Y}_i = \beta_{0i} + \beta_1 \bar{X}_{1i} + \bar{\epsilon}_i$$

and thus

$$Y_{ij} - \bar{Y}_i = \beta_1 (X_{1ij} - \bar{X}_{1i}) + (\epsilon_{ij} - \bar{\epsilon}_i)$$

- **Implementation:** create a new dataset, say, *newdata1*, with

$$Y_{ij}^* = Y_{ij} - \bar{Y}_i \quad \text{and} \quad X_{ij}^* = X_{1ij} - \bar{X}_{1i}$$

and then run a simple linear regression

$$Y^* = \beta_1 X^* + \epsilon^*$$

via the R code: `lm(Y* ~ X* - 1, newdata1)`

2. Adding binary Indicator variables

“Adding binary Indicator variable” for each subject

- **Key Ideas:**

$$\begin{aligned} Y_{ij} &= \beta_{0i} + \beta_1 X_{1ij} + \epsilon_{ij} \\ &= \beta_{01} + (\beta_{0i} - \beta_{01}) + \beta_1 X_{1ij} + \epsilon_{ij} \end{aligned}$$

- **Implementation:** set the “ID” variable in the original dataset, say, *data0*, as a factor variable, and then run a simple linear regression

$$Y_{ij} = \beta'_0 + \beta'_2 * ID_i + \beta'_1 X_{1ij} + \epsilon'_{ij}$$

via the R code: `lm(Y ~ factor(ID) + X, data0)`

R demo

- R demo for fixed-effects models